

```

1 # 0) GPU + repo
2 !nvidia-smi
3 %cd /content
4 !git clone -b hamza/discrim https://github.com/1hamzaibqbal/MFCLIP_acv
5 %cd MFCLIP_acv

```

```

1 # 1) Deps
2 !pip install torch torchvision timm einops yacs tqdm opencv-python scikit-learn scipy pyyaml ruamel.yaml pytorch
3
4

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1 # 2) Data (example: Oxford Pets into /content/data/oxford_pets)
2 from torchvision.datasets import OxfordIIITPet
3 from torchvision import transforms
4 from pathlib import Path
5 root = Path("/content/data/oxford_pets")
6 root.mkdir(parents=True, exist_ok=True)
7 _ = OxfordIIITPet(root=str(root), download=True, transform=transforms.ToTensor())
8 print("Oxford Pets downloaded to", root)
9
10

```

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1 # fetch official Oxford-IIIT Pet dataset (provides annotations/trainval.txt)
2 %cd /content
3 !mkdir -p /content/data/oxford_pets
4 !wget -q https://www.robots.ox.ac.uk/~vgg/data/pets/data/images.tar.gz
5 !wget -q https://www.robots.ox.ac.uk/~vgg/data/pets/data/annotations.tar.gz
6 !tar -xf images.tar.gz -C /content/data/oxford_pets
7 !tar -xf annotations.tar.gz -C /content/data/oxford_pets
8 !ls /content/data/oxford_pets/annotations | head
9

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1 # 3) Mount Drive and Setup Checkpoint
2 from google.colab import drive
3 import shutil
4 import os
5
6 drive.mount('/content/drive')
7
8 # Define paths
9 src_ckpt = "/content/drive/MyDrive/grad/comp_vision/hanson_loss/oxford_pets/RN50_ArcFace_oxford_pets.pth"
10 dst_dir = "/content/data/oxford_pets"
11 dst_ckpt = os.path.join(dst_dir, "RN50_ArcFace.pth")
12
13 # Copy checkpoint
14 os.makedirs(dst_dir, exist_ok=True)
15 if os.path.exists(src_ckpt):
16     shutil.copy(src_ckpt, dst_ckpt)
17     print(f"Copied {src_ckpt} to {dst_ckpt}")
18 else:
19     print(f"Warning: Checkpoint not found at {src_ckpt}. Please ensure Drive is mounted and path is correct.")
20

```

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1 # 4) Train ViT Generator
2 !python main.py \
3     --flag train_unet \
4     --generator vit \
5     --dataset oxford_pets \
6     --root /content/data \
7     --config-file configs/trainers/CoOp/rn50.yaml \
8     --dataset-config-file configs/datasets/oxford_pets.yaml \
9     --trainer ZeroshotCLIP \
10    --surrogate RN50 \
11    --head ArcFace \
12    --num_epoch 1 \
13    --bs 64 \
14    --lr 0.01 \
15    --optimizer SGD \
16    --ratio 0.2 \
17    --device cuda:0
18

```

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Epoch: 167, train acc: 0.0030, loss: -7.955196
Epoch: 168, train acc: 0.0030, loss: -7.684633
Epoch: 169, train acc: 0.0038, loss: -7.947381
Epoch: 170, train acc: 0.0049, loss: -8.104212
Epoch: 171, train acc: 0.0022, loss: -8.052837
Epoch: 172, train acc: 0.0033, loss: -7.883619
Epoch: 173, train acc: 0.0088, loss: -8.055849
Epoch: 174, train acc: 0.0033, loss: -8.154216
Epoch: 175, train acc: 0.0025, loss: -8.001501
Epoch: 176, train acc: 0.0038, loss: -8.015456
Epoch: 177, train acc: 0.0030, loss: -8.040501
Epoch: 178, train acc: 0.0027, loss: -8.024829
Epoch: 179, train acc: 0.0016, loss: -8.187323
Epoch: 180, train acc: 0.0022, loss: -8.110501
Epoch: 181, train acc: 0.0044, loss: -7.967625
Epoch: 182, train acc: 0.0044, loss: -8.085733
Epoch: 183, train acc: 0.0047, loss: -8.076750
Epoch: 184, train acc: 0.0060, loss: -8.205884
Epoch: 185, train acc: 0.0014, loss: -8.189281
Epoch: 186, train acc: 0.0019, loss: -8.169051
Epoch: 187, train acc: 0.0025, loss: -8.232175
Epoch: 188, train acc: 0.0011, loss: -8.345478
Epoch: 189, train acc: 0.0022, loss: -8.264975
Epoch: 190, train acc: 0.0027, loss: -8.116604
Epoch: 191, train acc: 0.0030, loss: -7.924865
Epoch: 192, train acc: 0.0033, loss: -8.092234
Epoch: 193, train acc: 0.0019, loss: -8.042344
Epoch: 194, train acc: 0.0044, loss: -8.105702
Epoch: 195, train acc: 0.0033, loss: -8.083128
Epoch: 196, train acc: 0.0044, loss: -8.261280
Epoch: 197, train acc: 0.0033, loss: -8.348648
Epoch: 198, train acc: 0.0019, loss: -8.287075
Epoch: 199, train acc: 0.0049, loss: -8.357465
Epoch: 200, train acc: 0.0019, loss: -8.451105
Epoch: 201, train acc: 0.0025, loss: -8.220890
Epoch: 202, train acc: 0.0082, loss: -8.382425
Epoch: 203, train acc: 0.0038, loss: -8.416080
Epoch: 204, train acc: 0.0016, loss: -8.452477
Epoch: 205, train acc: 0.0025, loss: -8.526575
Epoch: 206, train acc: 0.0022, loss: -8.504333
Epoch: 207, train acc: 0.0027, loss: -8.403243
Epoch: 208, train acc: 0.0016, loss: -8.378165
Epoch: 209, train acc: 0.0016, loss: -8.459722
Epoch: 210, train acc: 0.0008, loss: -8.429343
Epoch: 211, train acc: 0.0008, loss: -8.369009
Epoch: 212, train acc: 0.0036, loss: -8.420289
Epoch: 213, train acc: 0.0033, loss: -8.525370
Epoch: 214, train acc: 0.0016, loss: -8.476096
Epoch: 215, train acc: 0.0011, loss: -8.448888
Epoch: 216, train acc: 0.0025, loss: -8.551814
Epoch: 217, train acc: 0.0019, loss: -8.531777
Epoch: 218, train acc: 0.0030, loss: -8.608517
Epoch: 219, train acc: 0.0008, loss: -8.647182
Epoch: 220, train acc: 0.0022, loss: -8.594560
Epoch: 221, train acc: 0.0025, loss: -8.677957
Epoch: 222, train acc: 0.0011, loss: -8.627964
Epoch: 223, train acc: 0.0019, loss: -8.574197
Epoch: 224, train acc: 0.0019, loss: -8.543353

```

```

1 # 5) Save Artifacts and Plot History
2 import json
3 import matplotlib.pyplot as plt
4 import shutil
5 import os
6
7 # Paths
8 src_model = "/content/data/oxford_pets/vit_generator.pt"
9 src_history = "/content/data/oxford_pets/vit_generator_history.json"
10 dst_dir = "/content/drive/MyDrive/grad/comp_vision/hanson_loss/oxford_pets"
11
12 # Copy to Drive
13 os.makedirs(dst_dir, exist_ok=True)
14 if os.path.exists(src_model):
15     shutil.copy(src_model, os.path.join(dst_dir, "vit_generator.pt"))
16     print(f"Saved model to {dst_dir}/vit_generator.pt")
17 else:
18     print("Model file not found!")
19
20 if os.path.exists(src_history):
21     shutil.copy(src_history, os.path.join(dst_dir, "vit_generator_history.json"))
22     print(f"Saved history to {dst_dir}/vit_generator_history.json")
23

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24 # Plot
25 with open(src_history, 'r') as f:
26     history = json.load(f)
27
28 fig, ax1 = plt.subplots()
29
30 color = 'tab:red'
31 ax1.set_xlabel('Epoch')
32 ax1.set_ylabel('Loss', color=color)
33 ax1.plot(history['epoch'], history['loss'], color=color)
34 ax1.tick_params(axis='y', labelcolor=color)
35
36 ax2 = ax1.twinx()
37 color = 'tab:blue'
38 ax2.set_ylabel('Accuracy', color=color)
39 ax2.plot(history['epoch'], history['acc'], color=color)
40 ax2.tick_params(axis='y', labelcolor=color)
41
42 plt.title("Training Progress")
43 plt.show()
44 else:
45     print("History file not found!")
46
```

Saved model to /content/drive/MyDrive/grad/comp_vision/hanson_loss/oxford_pets/vit_generator.pt
History file not found!

```
1 from google.colab import drive
2 drive.mount('/content/drive')
```

Mounted at /content/drive

1 Start coding or generate with AI.